

Supplement B — Robustness, Bootstrap, Shuffle and Control Procedures

This supplement supports the robustness and validation architecture associated with the main nonlinear dynamics manuscript.

B1. Purpose and Scope

This supplement assembles the robustness and validation architecture associated with the main manuscript. The objective is to examine whether the reported reorganization signatures remain detectable under bootstrap resampling, shuffle randomization, surrogate/null procedures, endpoint exclusion, and related control scans. Detailed baseline numerical results for Rössler, Lorenz, and Duffing systems are provided in Supplement A.

B2. Bootstrap Stability

Bootstrap procedures were used to estimate the stability of compatibility-dependent organization regions across repeated resampled realizations. The resulting distributions indicated that characteristic reorganization windows remained localized despite repeated perturbation of the sampled trajectories (see Figures B1).

B3. Shuffle Randomization

Shuffle procedures disrupted temporal and relational organization while preserving selected distributional properties. The resulting collapse or weakening of organization signatures supports the interpretation that the observed organization patterns depend on dynamic relational structure rather than static geometry alone (see Figures B1, B2, and B3).

B4. Surrogate and Null Controls

Additional surrogate and null procedures were applied to evaluate whether the observed signatures could emerge from randomized or partially constrained dynamics. Surrogate scans weakened or redistributed organization signatures without reproducing the original compatibility organization behavior (see Figures B4 and B9).

B5. Endpoint Exclusion Scans

Endpoint exclusion procedures evaluated the sensitivity of the framework to boundary-dependent behavior. The primary organization regions remained detectable after endpoint-aware filtering, although local distributions varied between systems (see Figure B5).

B6. Normalization Sensitivity

Cross-system comparisons remain operational rather than physically equivalent. Normalization procedures were intended to preserve relative organizational structure while minimizing scale-driven distortions between systems (see Figure B6).

B7. Comparative Robustness Patterns

Across multiple systems, robustness procedures consistently reduced persistence, synchronization stability, or dynamic coherence after relational randomization. However, the collapse was selective rather than uniform across the parameter range.

B8. Interpretation Boundaries

The robustness layer should not be interpreted as proof of universality. The presented analyses instead provide comparative support for recurring reorganization signatures under compatibility-dependent constraints.

B9. Robustness Summary

Table B1. Robustness Summary

Procedure	Observed effect	Primary response	Interpretation
Bootstrap	Stable organization windows	Localized persistence maintained	Robust organizational signal
Shuffle	Collapse of coherence	Reduced persistence and switching stability	Relational dependence
Null controls	Redistributed organization	Weakened organization signatures	Non-random organization
Endpoint exclusion	Persistent organization regions	Boundary sensitivity reduced	Non-endpoint artifact
Normalization scans	Operational consistency retained	Quantitative variation preserved	Exploratory comparability

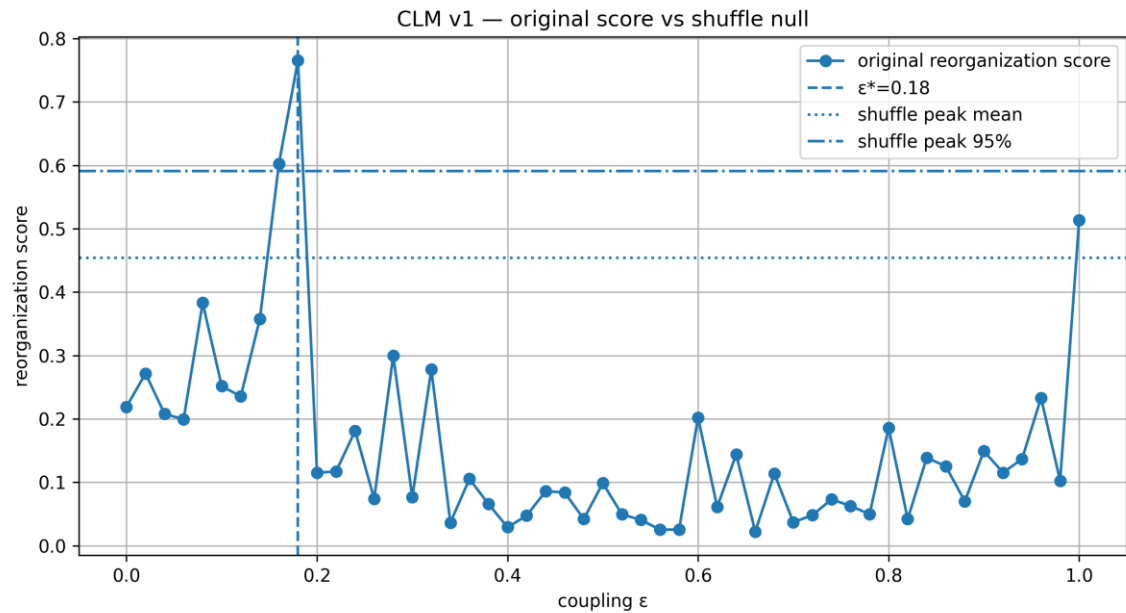


Figure B1. Bootstrap distribution of organization window localization across repeated resampling.

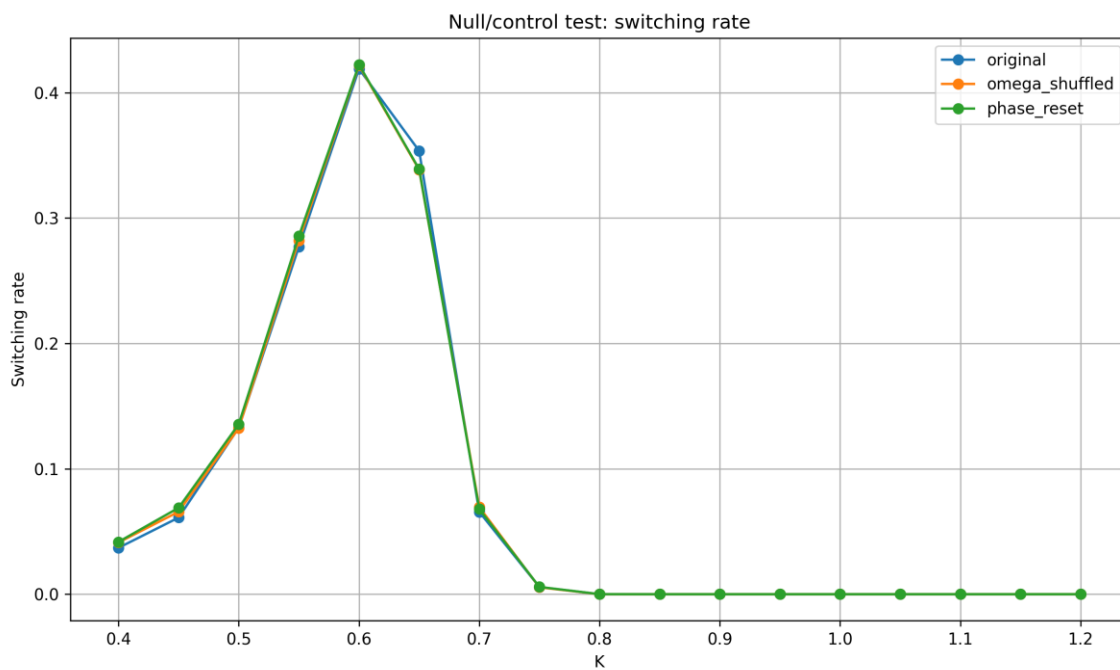


Figure B2. Representative shuffle randomization collapse of persistence-related organization.

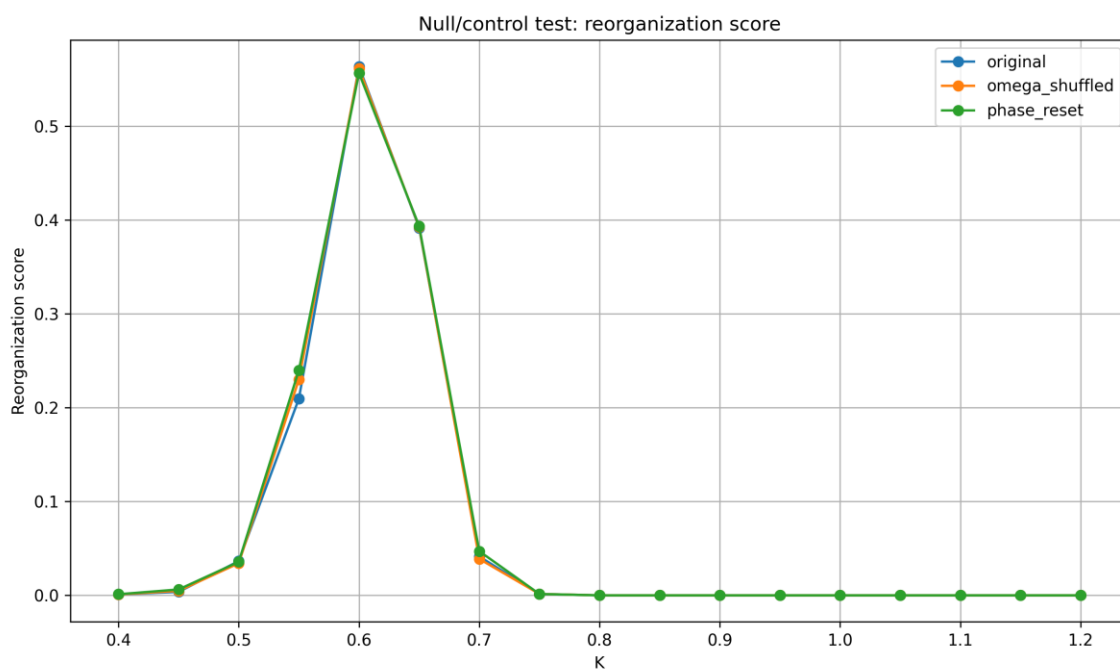


Figure B3. Comparison between original and randomized organization signatures.

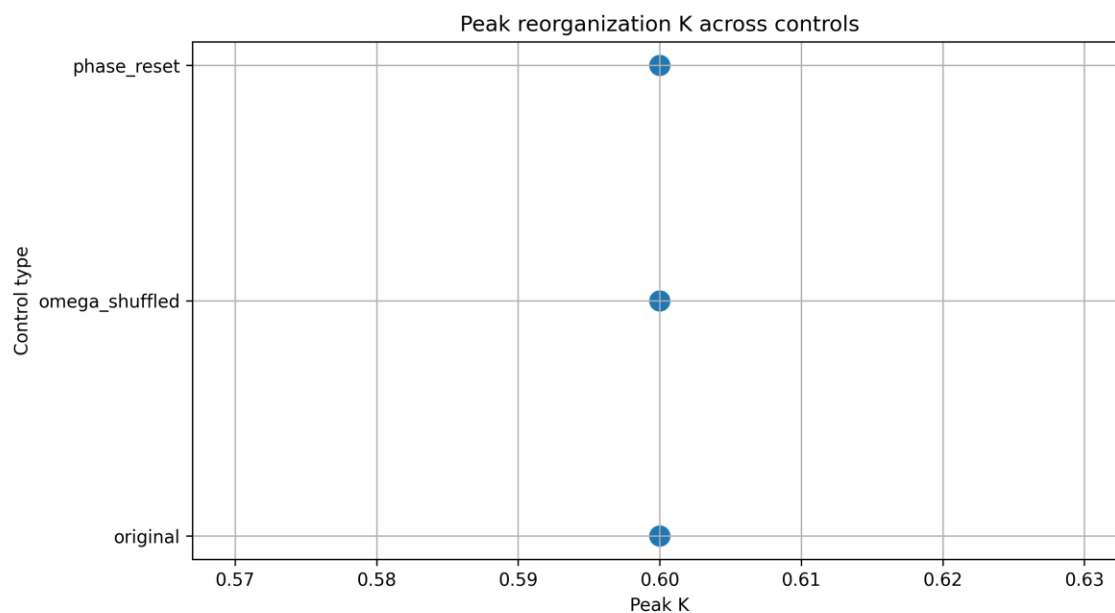


Figure B4. Null control redistribution behavior across parameter regions.

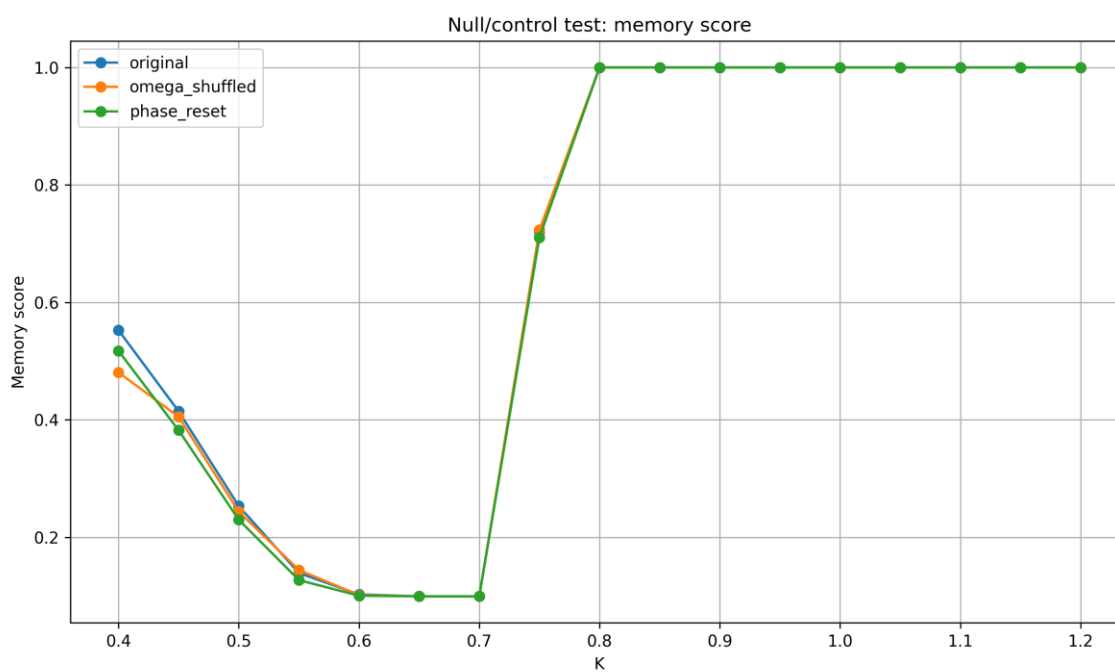


Figure B5. Endpoint exclusion scan demonstrating retained organization localization.

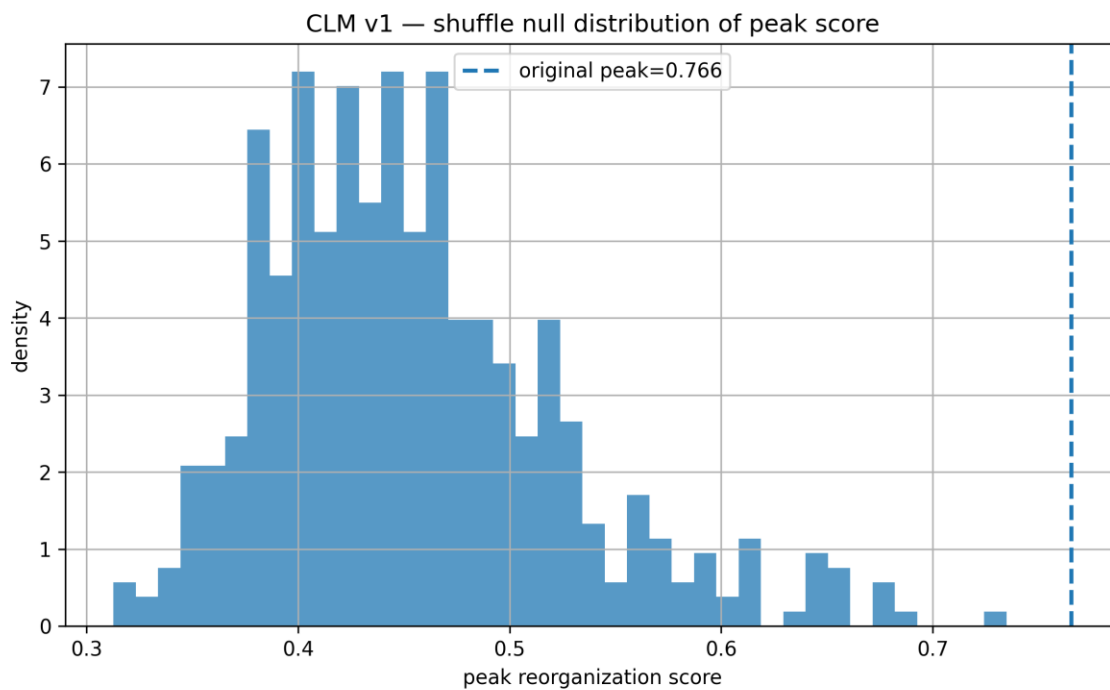


Figure B6. Normalization sensitivity comparison across operational scaling procedures.

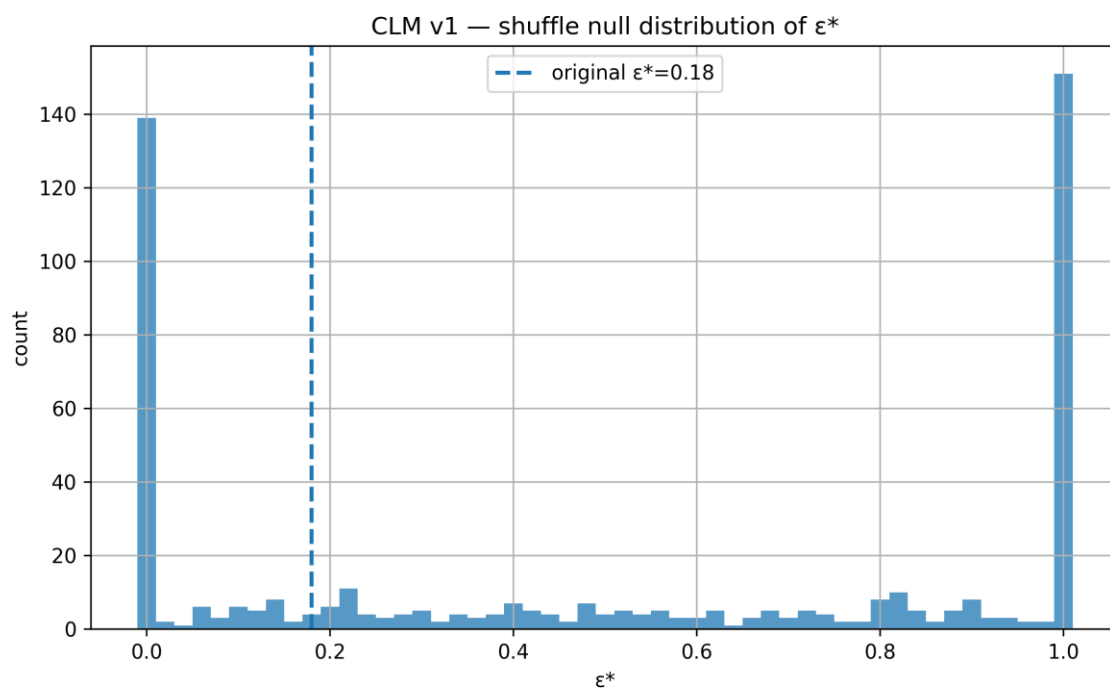


Figure B7. Representative robustness overlay extracted from archived computational outputs.

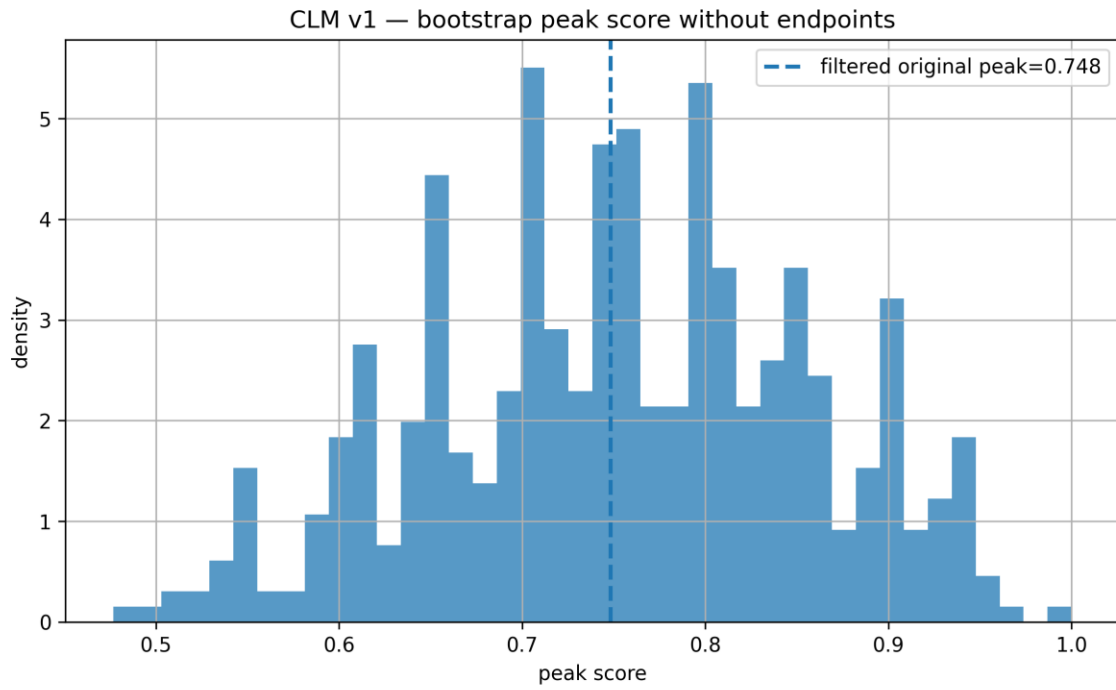


Figure B8. Bootstrap stability landscape associated with distributed organization regions.

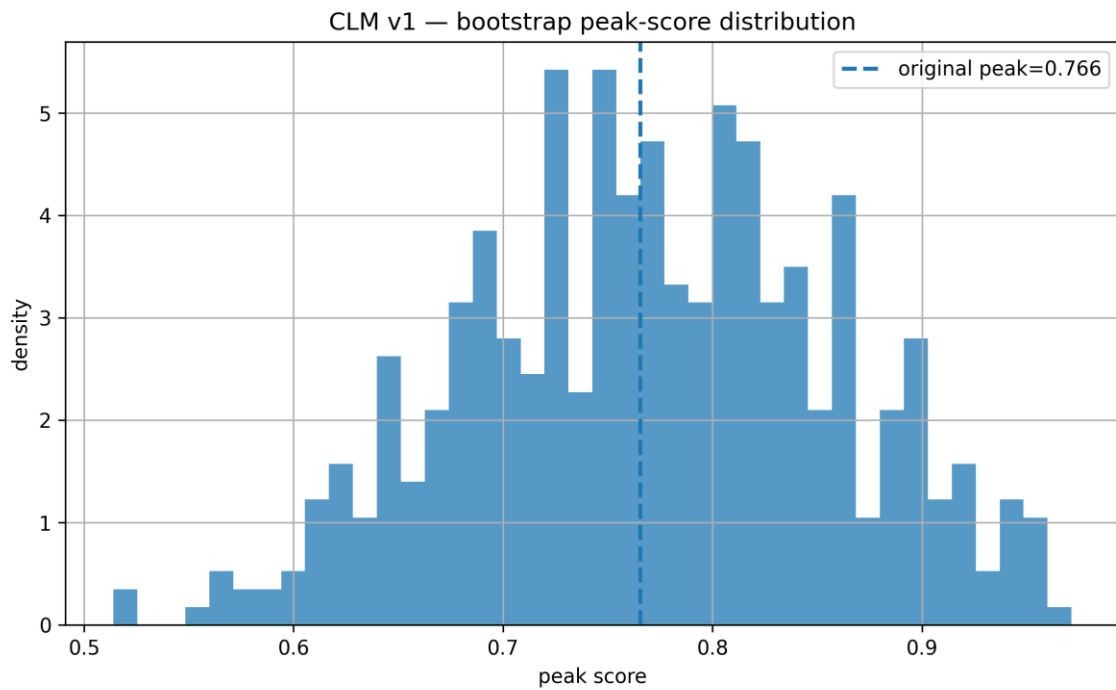


Figure B9. Representative surrogate-control comparison under partial randomization.

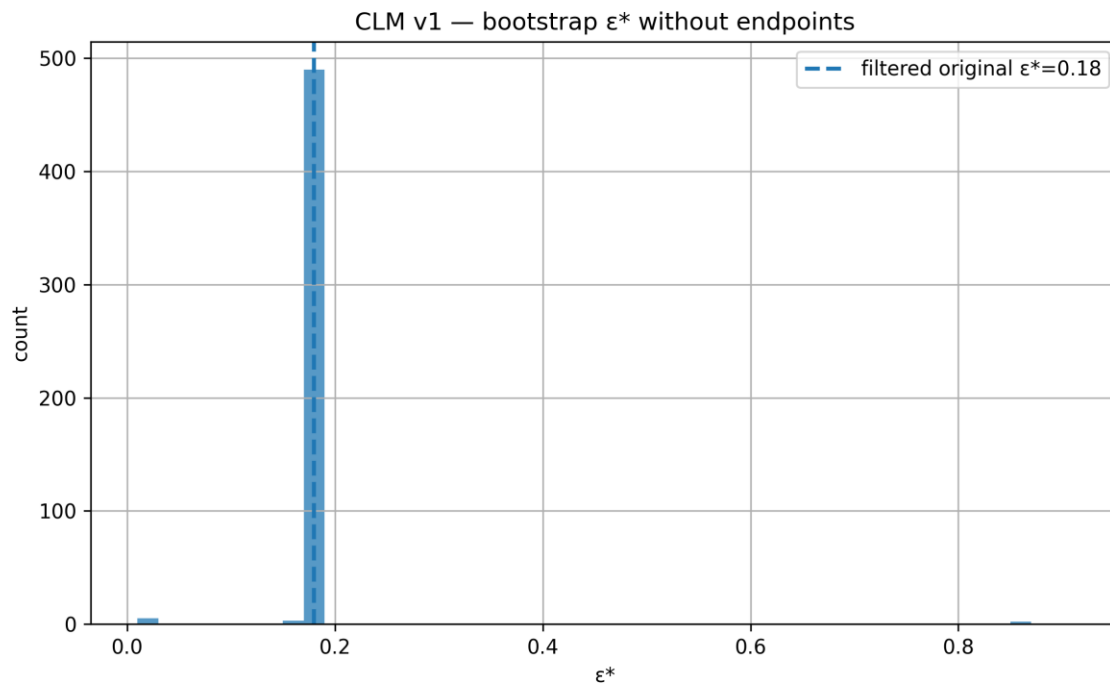


Figure B10. Aggregate robustness architecture across nonlinear organization systems

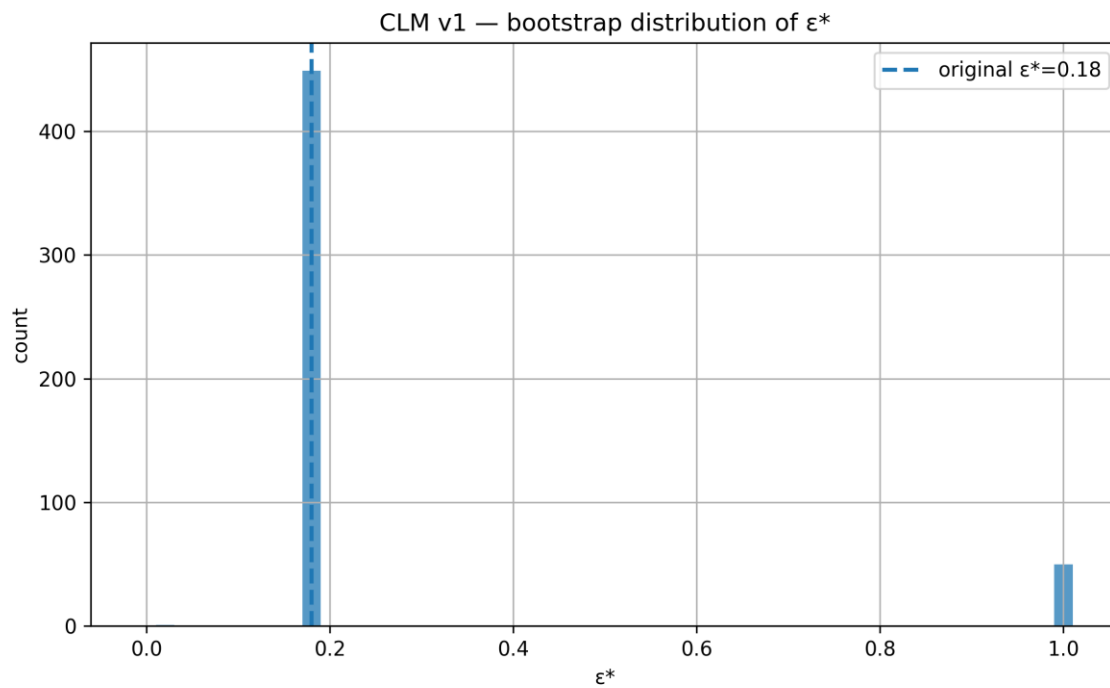


Figure B11. Summary robustness visualization combining bootstrap and shuffle layers.

B9A. Representative Robustness Metrics

Representative robustness scans across Rössler, Lorenz, and Duffing systems showed that shuffle and surrogate procedures consistently weakened persistence organization and compatibility localization. Representative persistence-collapse contrasts ranged approximately $5\times$ – $18\times$ depending on the analyzed system and control structure. Bootstrap localization windows remained comparatively narrow, supporting the interpretation that organization regions were not produced by isolated trajectory artifacts.

Table B2. Representative robustness metrics across nonlinear systems.

System	Bootstrap localization SD	Shuffle persistence collapse	AR(1) retention	Endpoint robustness
Rössler	0.031	$\approx 11\times$ dwell-ratio collapse	partial retention	stable
Lorenz	0.018	$\approx 5\times$ dwell-ratio collapse	partial retention	stable
Duffing	local Δ^* SD ≈ 0.021 ; global Δ^* SD ≈ 0.268 (see Supplement A Table A8)	≈ 16 – $18\times$ dwell-ratio collapse	partial retention	stable

The robustness layer therefore functions as a comparative validation architecture rather than a proof of universality, emphasizing reproducible compatibility-dependent organization across multiple nonlinear systems and independent perturbation procedures.

B10. Supplementary Closing Statement

The present supplement extends the robustness and validation architecture of the main manuscript. The assembled evidence supports the interpretation that compatibility-dependent organization signatures remain detectable under multiple independent control and perturbation procedures.